STORM WATCHER

Weather Journal

Watch it. Measure it. Predict it. Stay safe.



SPECIMEN LABEL

Meteorologist: _____

Journal started: _____ **Days logged:** _____

If found, please return to the DDMES Science Olympiad binder shelf.

How a storm watcher works

Scientists who study weather are called **meteorologists** (mee-tee-or-OL-oh-jists). Real storm chasers are trained adults. **Our** storm chasing happens from a safe window, with tools, maps, and data.

1
LOOK UP
Watch the sky at the same time each day. Name the clouds.

2
MEASURE
Temperature, rain, wind direction and speed, air pressure.

3
RECORD
Write it in your Weather Log right away. Scientists never trust memory!

4
PREDICT
Guess tomorrow's weather, then check if you were right.

5
STAY SAFE
Know the plan for every kind of storm.

The Field Scientist Code

- [ ] When thunder roars, go indoors. Wait 30 minutes after the last thunder.
- [ ] Watch storms only from inside, away from windows during strong wind.
- [ ] Turn Around, Don't Drown: never walk, bike, or ride through flood water.
- [ ] During a tornado warning, go to the lowest floor, in a small inside room.
- [ ] Never touch downed power lines or anything touching them.
- [ ] Weather apps and TV forecasts are tools too. Ask an adult to help.

My field kit

- [ ] This journal and a pencil
- [ ] Outdoor thermometer
- [ ] Rain gauge (a clear straight jar and a ruler works)
- [ ] Wind vane or a ribbon on a stick
- [ ] A compass (or know where the sun rises: east)
- [ ] A weather app or the TV forecast (with an adult)

Journal map

Tools and guides	p. 3-8
Sky Watch pages	p. 9-12
Hunts and missions	p. 13-18
Storm Log	p. 19

Where to look

Out a window
clouds, rain, wind in the trees, hail, lightning (from inside!)

Your backyard
rain gauge, thermometer, puddles, erosion spots

Weather apps and TV
radar, fronts, H and L, watches and warnings

Creeks after rain
fast brown water carrying soil (look from far away)

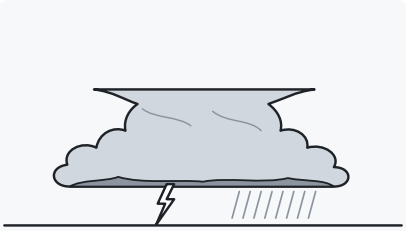
Beaches and rivers
storm damage and erosion from past hurricanes

Family stories
big storms that people you know remember

Did you know? Lightning is about 5 times hotter than the surface of the Sun, and Hurricane Florence dropped almost 36 inches of rain on one NC town in 2018!

The 5 kinds of severe weather

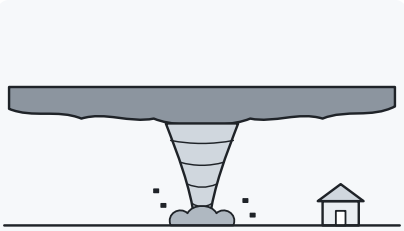
Check the box and write the date when you see (or live through) one. News and videos count as seeing one too.



Thunderstorm

- Has lightning and thunder
- Tall cloud called a cumulonimbus
- Most common in summer afternoons

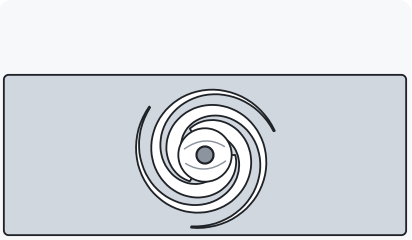
☐ Saw one: _____



Tornado

- Spinning column of air touching the ground
- Forms from a thunderstorm
- Usually lasts only minutes

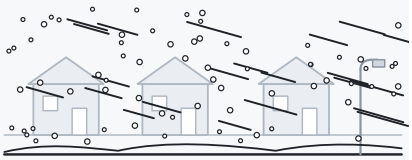
☐ Saw one: _____



Hurricane

- Forms over warm ocean water
- Has a calm eye in the middle
- Winds of 74 mph or more

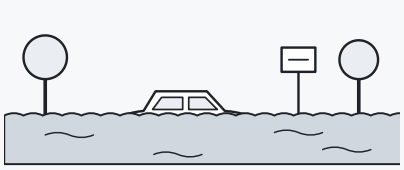
☐ Saw one: _____



Blizzard

- Snow with winds of 35 mph or more
- So hard to see you cannot see 1/4 mile
- Lasts 3 hours or more

☐ Saw one: _____



Flood

- Water covers land that is usually dry
- Flash floods happen in minutes
- Caused by heavy rain or ocean water pushed ashore

☐ Saw one: _____

Name that storm

Thunderstorm: lightning and thunder.

Tornado: spinning, touches the ground.

Hurricane: an eye, warm ocean.

Blizzard: snow + wind 35 mph+.

Flood: water where it's usually dry.

Weather tools

Meteorologists measure the air with these tools. Check **used** when you use one and **saw** when you see one (at school, on TV, or at the museum).

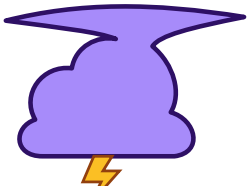
	Tool	What it measures	Me
	**Thermometer**	temperature (how hot or cold)	☐ used ☐ saw
	**Barometer**	air pressure (falling pressure often means storms)	☐ used ☐ saw
	**Anemometer**	wind speed	☐ used ☐ saw
	**Wind vane**	wind direction (it points to where the wind comes FROM)	☐ used ☐ saw
	**Rain gauge**	how much rain fell	☐ used ☐ saw
	**Radar**	where rain and storms are, and how heavy	☐ used ☐ saw
	**Satellite**	clouds and hurricanes from space	☐ used ☐ saw
	**Weather balloon**	temperature, wind, and moisture high up	☐ used ☐ saw

Make a rain gauge

1. Find a clear jar or bottle with straight sides.
2. Tape a ruler on the outside with 0 at the bottom.
3. Set it in an open spot away from roofs and trees.
4. Read it at the same time every day, then empty it.

Cloud Spotter

Clouds are forecasts you can read. Spot all 5 and write the dates you saw each one to earn the Cloud Spotter badge.

Cloud	What it looks like	Weather it brings	Dates I saw it
	**Cumulus** *KYOO-myoo-lus* Puffy cotton balls with flat bottoms.	Fair weather. If they grow tall, storms may come.	
	**Stratus** *STRAT-us* A flat gray blanket, low in the sky.	Gray days, drizzle, or fog.	
	**Cirrus** *SEER-us* Thin, wispy streaks very high up (made of ice).	Fair now, but a change may come in a day or two.	
	**Cumulonimbus** *kyoo-myoo-loh-NIM-bus* Huge, tall tower. The top can spread out like an anvil.	THUNDERSTORMS: lightning, hail, heavy rain, tornadoes.	
	**Nimbostratus** *nim-boh-STRAT-us* Thick, dark gray layer that hides the Sun.	Steady rain or snow for hours.	

Cloud names: *cumulus* = heap, *stratus* = layer, *cirrus* = curl of hair, *nimbus* = rain. So a cumulonimbus is a rain heap!

Weather Station Log: week 1

Use with Independent Lab I1 (Weather Station Week). Measure at the same time every day. Sky: clear, partly cloudy, cloudy, rain, snow, fog. Wind: use the Beaufort number from page 8. Pressure: rising, falling, or steady (from a weather app).

Date	Temp °F	Sky	Clouds	Wind from	Wind #	Rain (in)	Pressure	Tomorrow I predict… (✓ if right)

Graph the temperature: draw a dot for each day and connect them

This week's pattern

What happened to the pressure before it rained? Which day was warmest?

Weather Station Log: week 2

Use with Independent Lab I1 (Weather Station Week). Measure at the same time every day. Sky: clear, partly cloudy, cloudy, rain, snow, fog. Wind: use the Beaufort number from page 8. Pressure: rising, falling, or steady (from a weather app).

Date	Temp °F	Sky	Clouds	Wind from	Wind #	Rain (in)	Pressure	Tomorrow I predict… (✓ if right)

Graph the temperature: draw a dot for each day and connect them

This week's pattern

What happened to the pressure before it rained? Which day was warmest?

Wind Watcher and Lightning Counter

The Beaufort wind scale

No anemometer? Look at trees, flags, and smoke to estimate wind speed.

#	Name	What you see	mph
0	**Calm**	Smoke rises straight up	under 1
1	**Light air**	Smoke drifts; leaves still	1-3
2	**Light breeze**	Leaves rustle; you feel it on your face	4-7
3	**Gentle breeze**	Leaves and twigs move; flags flap	8-12
4	**Moderate breeze**	Dust and paper blow; small branches move	13-18
5	**Fresh breeze**	Small trees sway	19-24
6	**Strong breeze**	Big branches move; umbrellas are hard to hold	25-31
7	**Near gale**	Whole trees sway; hard to walk into the wind	32-38
8	**Gale**	Twigs break off trees	39-46

Flash to bang

Light travels much faster than sound. When you see lightning (from **inside**), count the seconds until you hear thunder. Divide by 5 to find how many **miles** away it was.

Date	Seconds	÷ 5 = miles

Is the storm coming closer?

If your count gets smaller each time, the storm is moving toward you. Stay inside until 30 minutes after the last thunder.

Safety: if you can hear thunder, you are close enough to be struck. Go inside!

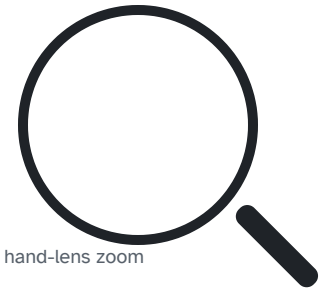
Sky Watch #1

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

Date _____ **Time** _____ **Place** _____

Sky: ☐ clear ☐ a few clouds ☐ half cloudy ☐ overcast ☐ foggy ☐ raining ☐ snowing

Sketch the sky: draw the clouds, and shade how much of the sky they cover.



hand-lens zoom

Real size: about _____ cm

CLOUDS I SEE

- ☐ cumulus
- ☐ stratus
- ☐ cirrus
- ☐ cumulonimbus
- ☐ nimbostratus
- ☐ none

MEASURE

Temp (°F) _____

Rain (in) _____

Wind from _____

Beaufort # _____

Pressure _____

WEATHER TODAY

_____

FRONT COMING?

_____

WATCH OR WARNING?

_____

TOMORROW I PREDICT

_____

WAS I RIGHT?

_____

Severe weather? ☐ none ☐ thunder ☐ lightning ☐ hail ☐ strong wind ☐ heavy rain ☐ snow or ice ☐ flooding

How did the weather affect people, plants, or animals?

_____

_____

_____

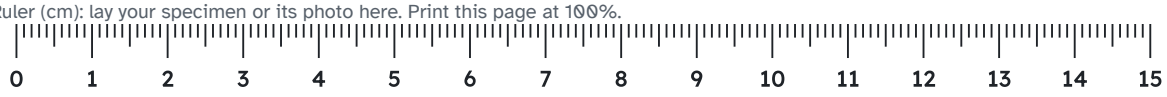
I wonder...

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Ruler (cm): lay your specimen or its photo here. Print this page at 100%.



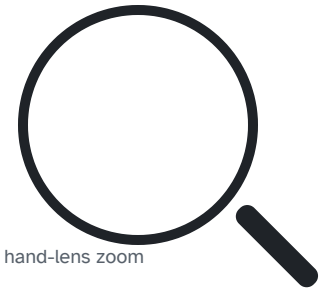
Sky Watch #2

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

Date _____ **Time** _____ **Place** _____

Sky: ☐ clear ☐ a few clouds ☐ half cloudy ☐ overcast ☐ foggy ☐ raining ☐ snowing

Sketch the sky: draw the clouds, and shade how much of the sky they cover.



hand-lens zoom

Real size: about _____ cm

CLOUDS I SEE

- ☐ cumulus
- ☐ stratus
- ☐ cirrus
- ☐ cumulonimbus
- ☐ nimbostratus
- ☐ none

MEASURE

Temp (°F) _____

Rain (in) _____

Wind from _____

Beaufort # _____

Pressure _____

WEATHER TODAY

_____

FRONT COMING?

_____

WATCH OR WARNING?

_____

TOMORROW I PREDICT

_____

WAS I RIGHT?

_____

Severe weather? ☐ none ☐ thunder ☐ lightning ☐ hail ☐ strong wind ☐ heavy rain ☐ snow or ice ☐ flooding

How did the weather affect people, plants, or animals?

_____

_____

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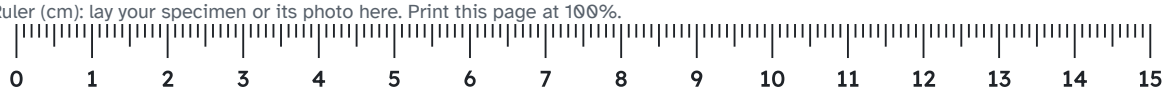
I wonder...

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Ruler (cm): lay your specimen or its photo here. Print this page at 100%.



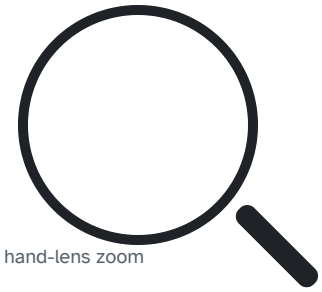
Sky Watch #3

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

Date _____ **Time** _____ **Place** _____

Sky: ☐ clear ☐ a few clouds ☐ half cloudy ☐ overcast ☐ foggy ☐ raining ☐ snowing

Sketch the sky: draw the clouds, and shade how much of the sky they cover.



hand-lens zoom

Real size: about _____ cm

CLOUDS I SEE

- ☐ cumulus
- ☐ stratus
- ☐ cirrus
- ☐ cumulonimbus
- ☐ nimbostratus
- ☐ none

MEASURE

Temp (°F) _____

Rain (in) _____

Wind from _____

Beaufort # _____

Pressure _____

WEATHER TODAY

_____

FRONT COMING?

_____

WATCH OR WARNING?

_____

TOMORROW I PREDICT

_____

WAS I RIGHT?

_____

Severe weather? ☐ none ☐ thunder ☐ lightning ☐ hail ☐ strong wind ☐ heavy rain ☐ snow or ice ☐ flooding

How did the weather affect people, plants, or animals?

_____

_____

_____

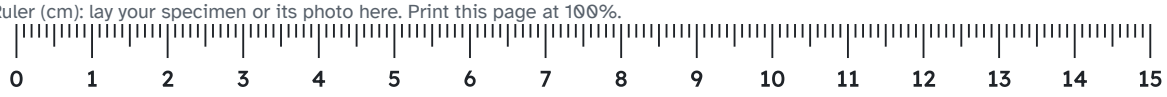
I wonder...

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Ruler (cm): lay your specimen or its photo here. Print this page at 100%.



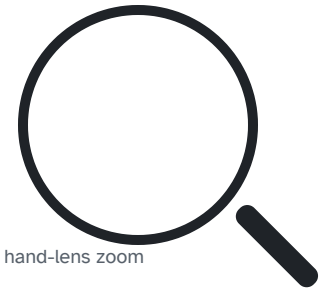
Sky Watch #4

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

Date _____ **Time** _____ **Place** _____

Sky: ☐ clear ☐ a few clouds ☐ half cloudy ☐ overcast ☐ foggy ☐ raining ☐ snowing

Sketch the sky: draw the clouds, and shade how much of the sky they cover.



hand-lens zoom

Real size: about _____ cm

CLOUDS I SEE

- ☐ cumulus
- ☐ stratus
- ☐ cirrus
- ☐ cumulonimbus
- ☐ nimbostratus
- ☐ none

MEASURE

Temp (°F) _____

Rain (in) _____

Wind from _____

Beaufort # _____

Pressure _____

WEATHER TODAY

_____

FRONT COMING?

_____

WATCH OR WARNING?

_____

TOMORROW I PREDICT

_____

WAS I RIGHT?

_____

Severe weather? ☐ none ☐ thunder ☐ lightning ☐ hail ☐ strong wind ☐ heavy rain ☐ snow or ice ☐ flooding

How did the weather affect people, plants, or animals?

_____

_____

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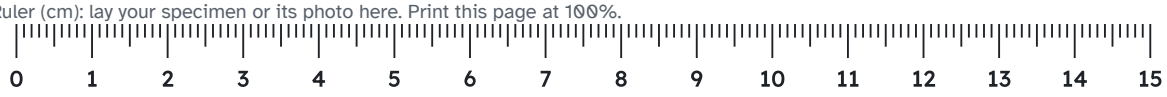
I wonder...

_____

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Ruler (cm): lay your specimen or its photo here. Print this page at 100%.



Storm Story Interview

Mission: interview a parent, grandparent, or neighbor about a big storm they remember. Scientists call this an **eyewitness report**.

Date _____ **Person I interviewed (relation, like "grandpa")** _____

1. What storm do you remember? (hurricane, tornado, flood, blizzard, big thunderstorm)

2. When and where did it happen?

3. What did the storm do? What did you see and hear?

4. How did you stay safe? What did you do before, during, and after?

5. What would you do differently next time?

Draw a picture of their storm

Weather Map Reader

Weather maps use symbols. Learn them, then copy a real map from the TV news or a weather app.



Cold front: cold air pushes in. Storms along the line!



Warm front: warm air slides in. Clouds and steady rain.

H **High pressure: fair, calm weather**

L **Low pressure: clouds, storms**

Radar colors: **green** = light rain, **yellow** = moderate, **red** = heavy rain and storms, **purple** = very severe (maybe hail).

Copy today's weather map: draw the fronts, H and L, and where it is raining

Date: _____ Source: _____

What weather will North Carolina get tomorrow? How do you know?

Watch or Warning?

WATCH = get ready

The weather is right for a storm to form. Keep watching and have your plan ready.

WARNING = take action now

The storm is happening or about to happen. Go to your safe place now!

Mission: all season, listen for real watches and warnings on the news, a weather app, or a phone alert. Record each one and what your family did.

Date	Type (tornado, flood…)	Watch or warning?	Where	What we did

Our safe places

Tornado	
Hurricane	
Flood	
Thunderstorm	

Your family storm plan (Independent Lab I2) has the full details.

Erosion Detective

Mission: after a big rain, walk your yard, schoolyard, or a trail with an adult. Find a spot where water carried soil away (**erosion**) or dropped it somewhere new (**deposition**).

Date _____ **Place** _____ **Rain (in)** _____

What it looks like now

Where did the soil go? Draw arrows

Clues I found:

☐ bare soil ☐ little gullies (channels) ☐ exposed roots ☐ mud fan at the bottom ☐ muddy puddle
☐ sand washed onto a path ☐ a fallen tree

What caused it?

_____
_____
_____

How could people stop it? (plants, mulch, rocks, gutters)

_____
_____
_____

Field Missions

One mission for each Storm Chasers meeting. When you finish, show a coach or your Team Captain for initials, then color in the badge.

When	Mission	Badge	Done	Initials
After Fri Oct 9 Lesson 1	Interview an adult about a storm they remember (p. 13).	Storm Storyteller	☐	
After Fri Nov 13 Lesson 2	Weather Station Week (Lab I1): log 7 days on p. 6.	Weather Watcher	☐	
Nov to Jan Anytime	Spot all 5 cloud types and write the dates (p. 5).	Cloud Spotter	☐	
After Fri Jan 8 Lesson 3	Use the Beaufort scale on 5 days and teach someone the eye vs. the eyewall.	Wind Watcher	☐	
After Fri Feb 5 Lesson 4	Family storm plan (Lab I2) and fill in Our safe places (p. 15).	Safety Captain	☐	
Fri Mar 5 Lesson 5	Copy a real weather map with a front, H and L (p. 14).	Map Reader	☐	
Anytime Bonus	Erosion Detective (p. 16), flash-to-bang from inside, 4 Sky Watch pages.	Erosion / Lightning / Field Scientist	☐	

Tournament day: Sat Mar 13, Chatham Regional

- ☐ 2 sharp pencils (no notes allowed!)
- ☐ A good night's sleep
- ☐ Your flashcards to review the night before

- **No notes:** quiz yourself with flashcards every other day.
- Read maps and data carefully: check the key first.
- Name the storm, then the hazard, then the safety step.

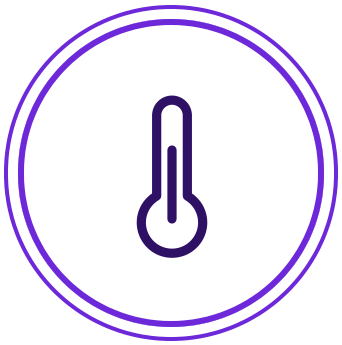
My Badges

Color in each badge when you earn it. Earn all 9 to become a **Master Storm Watcher**! Your Team Badge Passport has the same badges.



Storm Storyteller
Interview an adult about a storm

Initials: _____



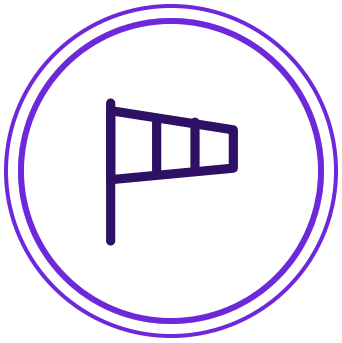
Weather Watcher
7 days in the Weather Log

Initials: _____



Cloud Spotter
Spot all 5 cloud types

Initials: _____



Wind Watcher
Beaufort scale on 5 days

Initials: _____



Safety Captain
Family storm plan + safe places

Initials: _____



Map Reader
Copy a real weather map

Initials: _____



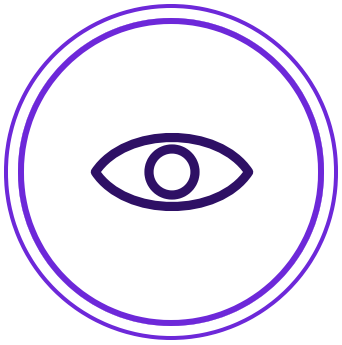
Erosion Detective
Find and explain erosion

Initials: _____



Lightning Counter
Flash-to-bang from inside

Initials: _____



Field Scientist
4 Sky Watch pages

Initials: _____

My Storm Log

Every real severe weather event you live through, or follow on the news, goes here.

#	Date	Weather event	Where	What it did	How we stayed safe
1					
2					
3					
4					
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